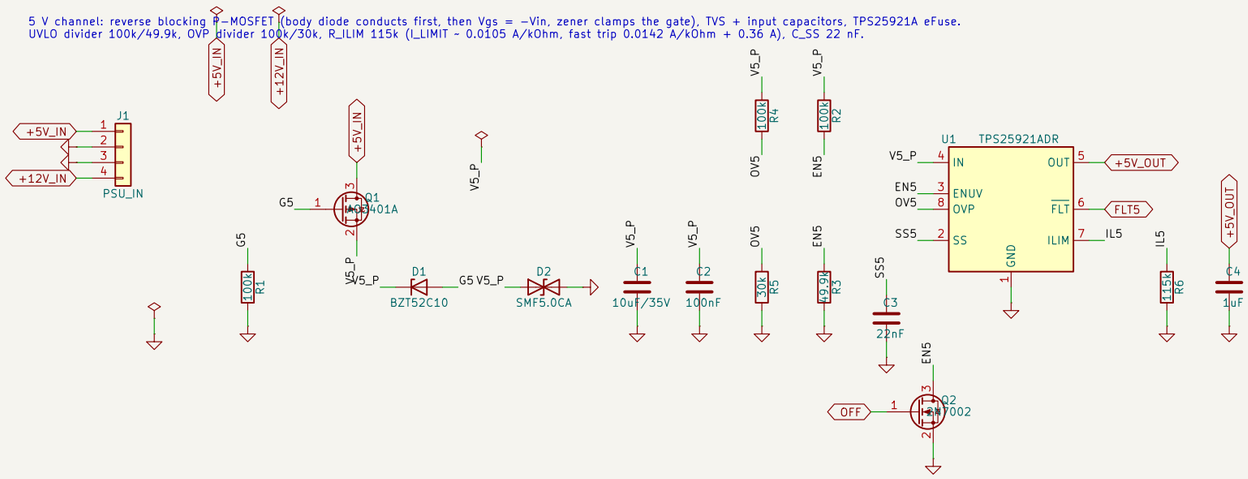
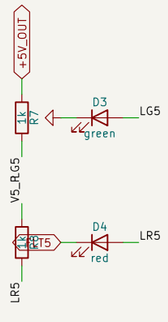
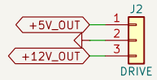
C64 PSU GUARD – protected 5 V / 12 V feed for a 1541-II / 1581 drive from a PC power supply.
J1 PSU in (Berg: 1 +5V, 2/3 GND, 4 +12V) -> J2 drive out (DIN pin 1 = 5V, 2 = GND, 4 = 12V).
Channel 5 V: limit 1.2 A, OVP 6.0 V, UVLO 4.2 V. Channel 12 V: limit 1.5 A, OVP 14.0 V, UVLO 10.4 V.
TPS25921A = auto retry: red LED on while the channel is off, off again when the fault is gone.
J3 UFI link (reserve): 1 GND, 2 OFF (high = both channels off), 3 FLT5, 4 FLT12 (open drain).

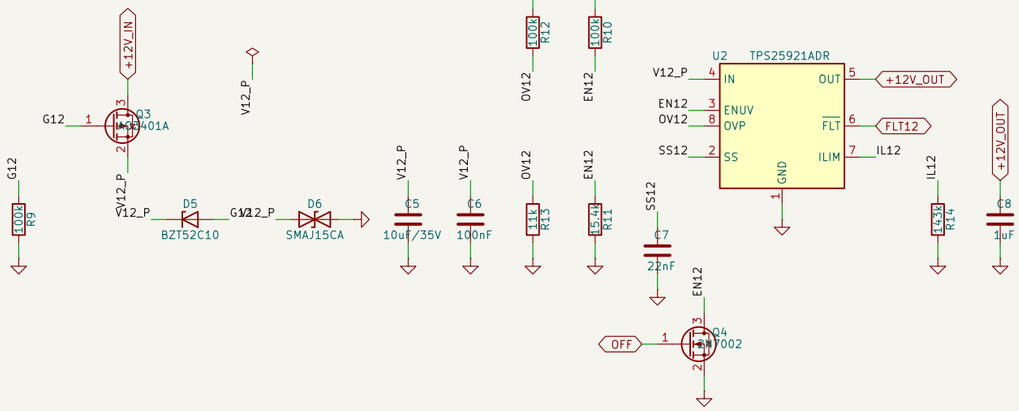
5 V channel: reverse blocking P-MOSFET (body diode conducts first, then Vgs = -Vin, zener clamps the gate), TVS + input capacitors, TPS25921A eFuse.
UVLO divider 100k/49.9k, OVP divider 100k/30k, R_ILIM 115k (I_LIMIT ~ 0.0105 A/kOhm, fast trip 0.0142 A/kOhm + 0.36 A), C_SS 22 nF.

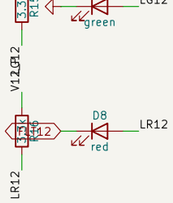




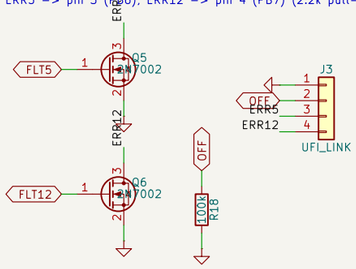


12 V channel: reverse blocking P-MOSFET (body diode conducts first, then Vgs = -Vin, zener clamps the gate), TVS + input capacitors, TPS25921A eFuse.
UVLO divider 100k/15.4k, OVP divider 100k/11k, R_ILIM 143k (I_LIMIT ~ 0.0105 A/kOhm, fast trip 0.0142 A/kOhm + 0.36 A), C_SS 22 nF.

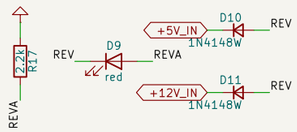




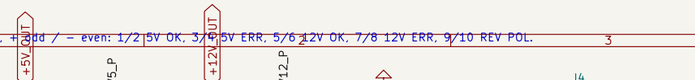
UFI link: OFF high = both channels off. ERR5/ERR12 = open drain, released (high) when the
channel reports a fault or the guard is unpowered, pulled low while the channel is on.
UFI Headless J9: ERR5 -> pin 3 (PB6), ERR12 -> pin 4 (PB7) (2.2k pull-ups there), OFF <- pin 5 (PE0), GND pin 9.



Reverse polarity LED: conducts only when +5V_IN or +12V_IN is negative against GND
(1N4148W blocks the rail voltage in normal operation).



J4 external LEDs, + odd / - even: 1/2 5V OK, 3/4 5V ERR, 5/6 12V OK, 7/8 12V ERR, 9/10 REV POL.



UFI Project
Sheet: /
File: C64_PSU_Guard.kicad_sch
Title: C64 PSU Guard
Size: A3 | Date: 2026-10-05 | **Rev: 0.1**
KiCad E.D.A. 10.0.6 | Id: 1/1